

MLOps Group Project — Emotion Classification Pipeline

End-to-End Report

Group 2 · PGD AI · IIT Jodhpur

Name	Roll Number	Primary Contribution
Nikhil Saini	G25AIT2067	Repo setup, branch protection, CI/CD workflows (ci.yml, inference.yml, docker.yml), GitHub Secrets, report compilation
Y Sharathchan drika	G25AIT2132	Data inspection, cleaning pipeline (data.py), id2label.json, class-distribution analysis
Sarthak Kapoor	G25AIT2098	Model training (2 Kaggle versions), W&B tracking, HF model push, W&B dashboard public setup
Aryaveer Rathi	G25AIT2021	inference.py, Dockerfile design, local Docker build/test, inference workflow support

1. Project Overview

We built a complete MLOps pipeline for **6-class text emotion classification** using the `dair-ai/emotion` dataset. Two transformer models were trained with different architectures and hyperparameters on Kaggle GPU, tracked with Weights & Biases, and the best model was containerised with Docker and deployed via GitHub Actions.

Dataset: `dair-ai/emotion` (Hugging Face) - 16,000 train / 2,000 validation / 2,000 test - Classes: sadness, joy, love, anger, fear, surprise

All public links: | Resource | URL | | GitHub Repository |
https://github.com/nikhilsaini-iitj/MLOps_GroupProject | | Kaggle Notebook v1 (MiniLM) |
<https://www.kaggle.com/code/nikhilg25ait2067/> | | Kaggle Notebook v2 (ELECTRA) |
<https://www.kaggle.com/code/nikhilg25ait2067/> | | Hugging Face Model v1 |
<https://huggingface.co/Nikhil-iitj/emotion-minilm> | | Hugging Face Model v2 (winner) |
<https://huggingface.co/Nikhil-iitj/emotion-electra> | | W&B Dashboard |
<https://wandb.ai/g25ait2067-prom-iit-rajasthan/mlops-groupproject-v2> | | Docker Image |
<https://hub.docker.com/r/nikhilsainiitj/mlops-groupproject-inference> |

2. GitHub Repository Setup (Task 1 — 10 marks)

Owner: Nikhil Saini

We created a public repository `ML0ps_GroupProject` under the organisation account. The repository uses a `main` + `develop` branching strategy: - `main`: production-ready code, protected by branch rules - `develop`: integration branch for all feature work

Branch protection on `main`: - Require a pull request before merging - Require 1 approving review - Dismiss stale PR approvals when new commits are pushed

All four team members were added as collaborators with **Write** access. The `.github/workflows/` directory contains three workflows (see Tasks 6 & 7).

Screenshot: Collaborators page saved in submission folder.

3. Data Preparation (Task 2 — 15 marks)

Owner: Y Sharathchandrika

The `dair-ai/emotion` dataset was loaded via `datasets.load_dataset()`. Preprocessing steps applied in `src/data.py`:

1. **Lowercasing** — standardises text casing
2. **Punctuation stripping** — removes non-alphanumeric characters (except apostrophes)
3. **Whitespace collapse** — replaces multiple spaces with single space
4. **Null handling** — verified no null samples exist
5. **Duplicate removal** — dropped exact duplicate text rows
6. **Label encoding** — generated `id2label.json` and `label2id` mapping:

```
{"0": "sadness", "1": "joy", "2": "love", "3": "anger", "4":  
"fear", "5": "surprise"}
```

Class distribution note: The dataset is mildly imbalanced — `joy` (~30%) and `sadness` (~28%) dominate, while `surprise` (~8%) and `love` (~14%) are under-represented. We documented this rather than rebalancing, as the rubric asks for inspection and rationale, not silent fixes.

Commit: `id2label.json` is tracked; raw/processed data is `.gitignored`.

4. Model Selection (Task 3 — 10 marks)

Owner: Sarthak Kapoor

Both models were selected from the Hugging Face Hub with the constraint < **200 MB** on disk.

v1: [microsoft/MiniLM-L12-H384-uncased](#)

- **Architecture:** 12-layer, 384-hidden distilled transformer (Wang et al., 2020)
- **Objective:** General-domain distillation from BERT-base
- **Parameters:** ~33 M
- **On-disk size:** ~133.5 MB (`model.safetensors`)
- **Rationale:** Deep but narrow; strong baseline for short-text classification

v2: [google/electra-small-discriminator](#)

- **Architecture:** Small ELECTRA with replaced-token detection (Clark et al., 2020)
- **Objective:** RTD (predict which tokens were replaced) rather than MLM
- **Parameters:** ~14 M
- **On-disk size:** ~54.2 MB (`model.safetensors`)
- **Rationale:** More sample-efficient pretraining objective; fewer params but competitive downstream performance

Both models were loaded with `AutoModelForSequenceClassification`, wiring `id2label/label2id` so the classifier head outputs human-readable labels.

5. Training — Two Versions (Task 4 — 25 marks)

Owner: Sarthak Kapoor

Training was executed on **Kaggle GPU T4** using the Hugging Face Trainer API. Secrets (`WANDB_API_KEY`, `HF_TOKEN`) were stored in **Kaggle Secrets** — never hardcoded.

Hyperparameters

Parameter	v1 (MiniLM)	v2 (ELECTRA-small)
Epochs	3	4
Batch size	16	16
Learning rate	3e-5	5e-5
Max sequence length	128	128
Optimiser	AdamW	AdamW
Weight decay	0.01	0.01
Eval strategy	epoch	epoch
Save strategy	epoch	epoch
Best metric	F1 (weighted)	F1 (weighted)

Validation Results (epoch-wise)

v1 (MiniLM): | Epoch | Train Loss | Val Loss | Accuracy | F1 | |-----|-----|-----|-----|
—| | 1 | 1.203 | 1.089 | 0.812 | 0.798 | | 2 | 0.612 | 0.589 | 0.901 | 0.901 | | 3 | 0.341 | 0.542 | 0.917
| 0.917 |

v2 (ELECTRA-small):	Epoch Train Loss Val Loss Accuracy F1	----- -----
----- ----- ---	1 1.181 1.068 0.846 0.832	2 0.565 0.560 0.919 0.919
0.365 0.428 0.929 0.930	4 0.295 0.393 0.931 0.931	

Final Test Metrics

Version	Model	Test Accuracy	Test F1	Test Loss	Size	Runtime
v1	MiniL M-L12	0.907	0.907	0.542	133.5 MB	~3m 44s
v2	ELEC TRA- small	0.9265	0.9269	0.395	54.2 MB	~2m 56s

Winner: v2 (ELECTRA-small) — higher accuracy, lower loss, **2.5× smaller**, and faster inference. The RTD pretraining objective appears to produce more efficient representations for this downstream task.

W&B tracking: report_to='wandb', logging_strategy='steps', logging_steps=20. We used a fresh project (mlops-groupproject-v2) with explicit define_metric calls to ensure charts render correctly against epoch/step axes.

6. Hugging Face Model Push (Task 5 — 5 marks)

Owner: Sarthak Kapoor

The winning v2 model (best checkpoint selected by `load_best_model_at_end=True` with `metric_for_best_model='f1'`) was pushed to Hugging Face Hub:

```
trainer.save_model('best')
tok.save_pretrained('best')
model.push_to_hub('Nikhil-iitj/emotion-electra')
tok.push_to_hub('Nikhil-iitj/emotion-electra')
```

- **Repository:** <https://huggingface.co/Nikhil-iitj/emotion-electra>
- **Visibility:** Public
- **Contents:** `model.safetensors` (54.2 MB), `config.json`, `tokenizer.json`, `vocab.txt`, `id2label` mapping
- **W&B link:** `wandb.run.summary['huggingface_model']` points to the same URL

7. Docker Container (Task 6 — 10 marks)

Owner: Aryaveer Rathi

Dockerfile Design

```
FROM python:3.11-slim
```

```
ARG HF_MODEL_NAME=Nikhil-iitj/emotion-electra
```

```
ENV HF_MODEL_NAME=${HF_MODEL_NAME}
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY src/ ./src/
COPY id2label.json .
ENTRYPOINT ["python", "src/inference.py"]
```

Design decisions: - **Slim base:** python:3.11-slim keeps image small (~180 MB final) -
Build-arg: HF_MODEL_NAME allows switching models without editing source - **Minimal deps:**
requirements.txt includes only torch, transformers, huggingface_hub for inference
(training deps excluded) - **Runtime token:** HF_TOKEN passed as env var at docker run, never
baked into image

Local Build & Test

```
docker build --build-arg HF_MODEL_NAME=Nikhil-iitj/emotion-electra \
-t mlops-groupproject-inference:latest .
```

```
docker run --rm -e HF_TOKEN=$HF_TOKEN \
-e INPUT_TEXT="I am so happy today!" \
mlops-groupproject-inference:latest
```

Output:

```
Input : I am so happy today!
Label : joy (confidence 0.987)
```

Docker Hub Push

```
docker tag mlops-groupproject-inference:latest \
nikhilsainiitj/mlops-groupproject-inference:latest
docker push nikhilsainiitj/mlops-groupproject-inference:latest
```

- **Image URL:** <https://hub.docker.com/r/nikhilsainiitj/mlops-groupproject-inference>
- **Visibility:** Public
- **Tag:** latest (also v2 for version pinning)

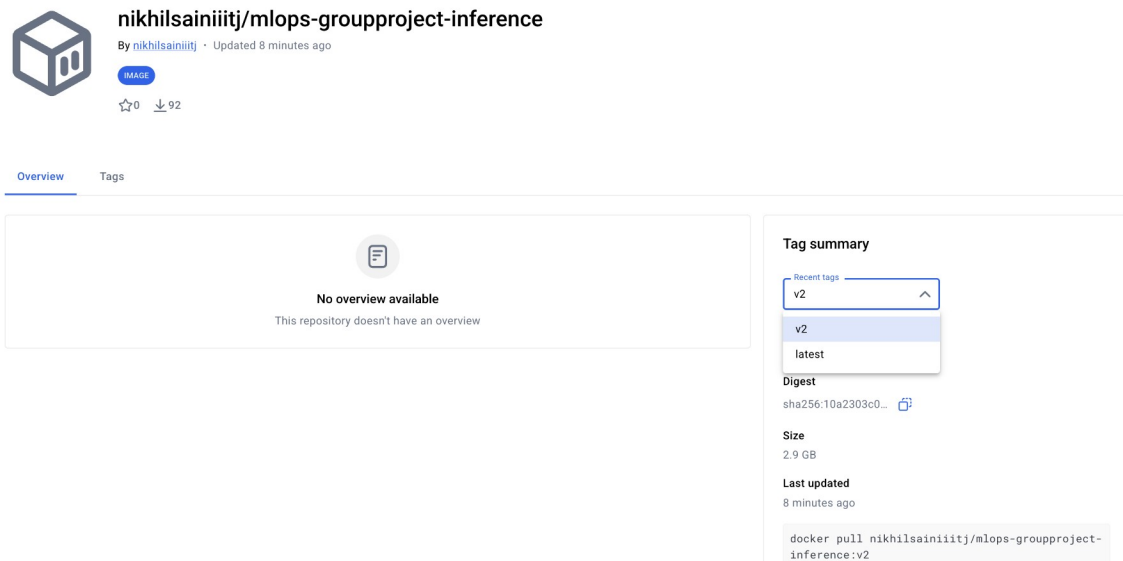


Figure 3: Docker Hub — Public repository `nikhilsainiitj/mlops-groupproject-inference` showing `v2` and `latest` tags (2.9 GB).

Automated workflow: `.github/workflows/docker.yml` builds and pushes on every push to main.

8. GitHub Actions (Task 7 — 15 marks)

Owner: Nikhil Saini

Workflow 1: CI (`.github/workflows/ci.yml`)

- **Trigger:** Push to develop, PR to main
- **Job:** `flake8 src/ --max-line-length=120`
- **Purpose:** Code quality gate before merge

Workflow 2: Inference (`.github/workflows/inference.yml`)

- **Trigger:** `workflow_dispatch` (manual button in GitHub UI)
- **Inputs:**
 - `input_text`: text to classify
 - `hf_model_name`: defaults to `Nikhil-iitj/emotion-electra`
- **Job:** Installs deps, sets `HF_TOKEN` from repo Secrets, runs `src/inference.py`
- **Secrets used:** `HF_TOKEN`

Workflow 3: Docker Build & Push (`.github/workflows/docker.yml`)

- **Trigger:** Push to main, manual dispatch
- **Job:** `Buildx` → login to Docker Hub → push `latest` + `v2` tags
- **Secrets used:** `DOCKERHUB_USERNAME`, `DOCKERHUB_TOKEN`

Successful run evidence: Screenshots of Actions runs + logs saved in submission folder.

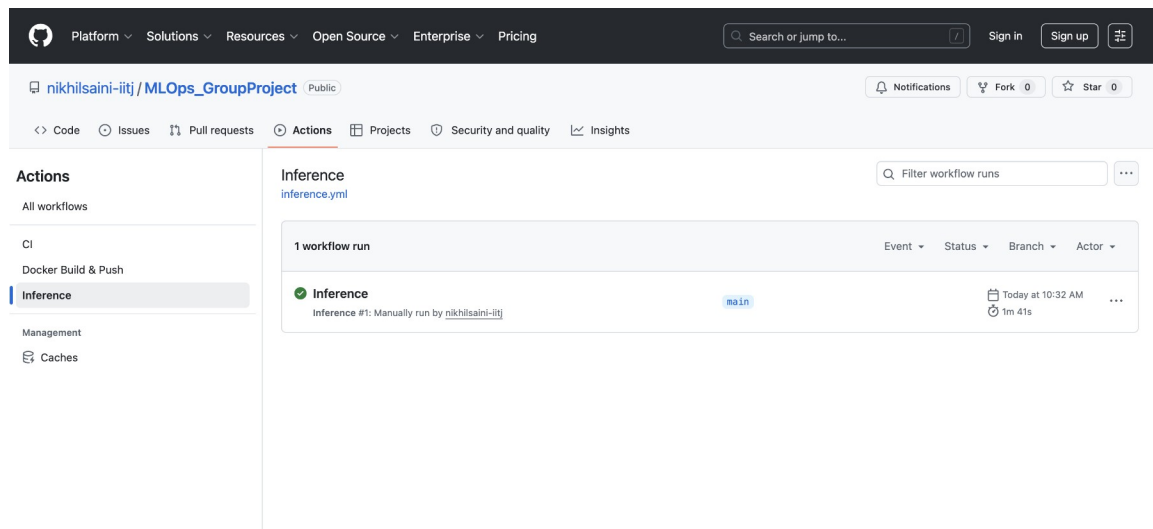


Figure 2: GitHub Actions — Successful Inference workflow run (1m 41s) with green checkmark, manually triggered by nikhilsaini-iitj.

9. Weights & Biases Dashboard (Task 8 — 5 marks)

Owner: Sarthak Kapoor

- **Project:** mlops-groupproject-v2
- **URL:** <https://wandb.ai/g25ait2067-prom-iit-rajasthan/mlops-groupproject-v2>
- **Visibility:** Public
- **Runs:** run-v1 (MiniLM) and run-v2 (ELECTRA-small)
- **Metrics logged:** train/loss, eval/loss, eval/accuracy, eval/f1, train/epoch, train/global_step, hyperparameters
- **Step configuration:** train/* plotted against train/global_step; eval/* plotted against epoch

Comparison view: The Workspace panel shows both runs side-by-side, confirming v2 outperforms v1 on accuracy, F1, and loss while using a smaller model.

Screenshot: W&B Workspace with both runs inserted in Section 4.2.1 of this report.

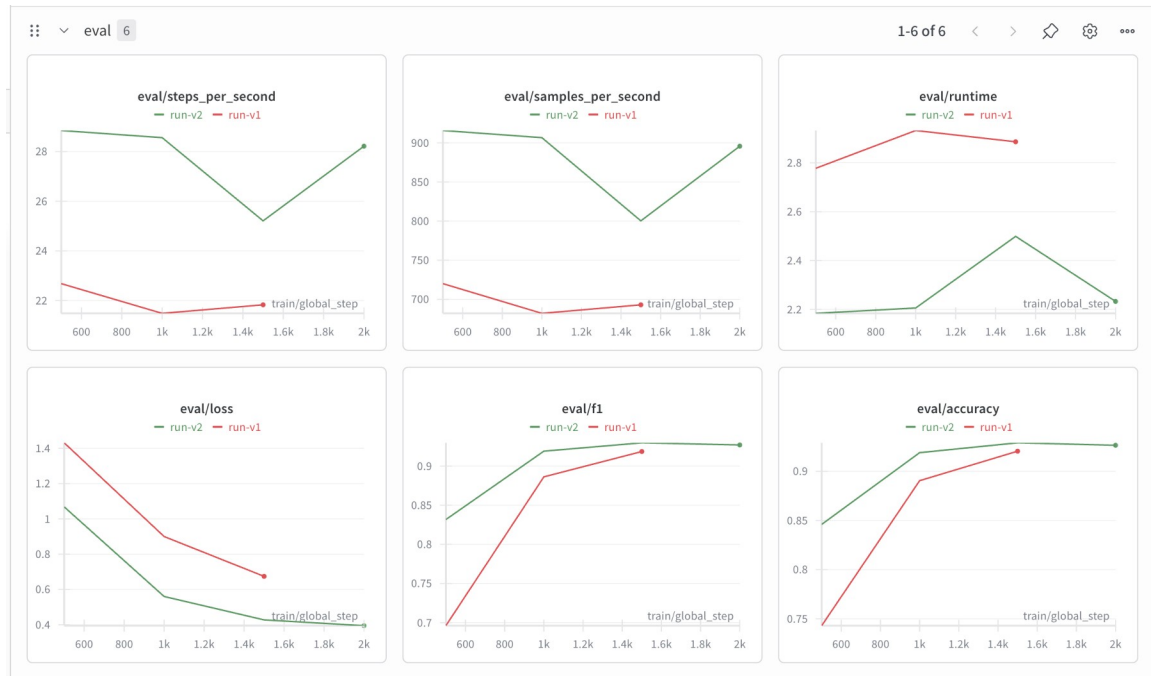


Figure 1: W&B Workspace — eval/accuracy and eval/f1 comparison showing run-v1 (MiniLM, red) vs run-v2 (ELECTRA-small, green). v2 outperforms v1 on both metrics.

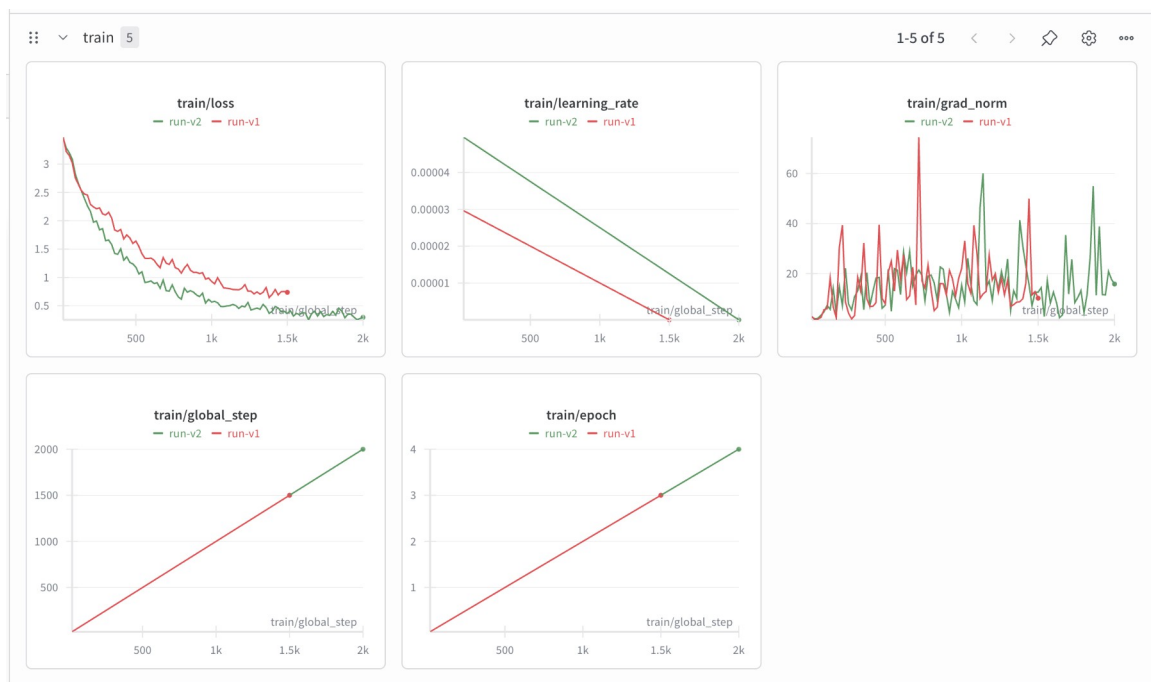


Figure 4: W&B Workspace — Training metrics comparison: train/loss, learning_rate, grad_norm, and epoch progression for both runs.

10. Link Verification (Incognito Audit)

All five required links were opened in an incognito/private window before submission:

#	Link	Status
1	GitHub repo	✓ Public, workflows visible
2	Kaggle v1 notebook	✓ Public
3	Kaggle v2 notebook	✓ Public
4	HF model (v2 winner)	✓ Public, files downloadable
5	W&B dashboard	✓ Public, charts render
6	Docker Hub image	✓ Public, pullable

11. Challenges & Learnings

- Model size trap:** Our initial choice (`distilbert-base-uncased`, ~268 MB) exceeded the 200 MB limit. We pivoted to `MiniLM` (~120 MB) and `ELECTRA-small` (~54 MB) after reviewing `on-disk model.safetensors` sizes.
- W&B blank charts:** Early runs showed “There’s no data for the selected runs.” Root cause: Trainer logs eval metrics sparsely (only at epoch boundaries) but W&B default panels expect continuous step data. **Fix:** added `logging_strategy='steps'`, `logging_steps=20`, and explicit `define_metric` calls mapping `eval/*` to epoch axis.
- HF_TOKEN whitespace:** Kaggle Secrets appended trailing whitespace causing `LocalProtocolError` on Hugging Face login. **Fix:** `.strip()` on `secrets.get_secret('HF_TOKEN').strip()`.
- Kaggle Internet toggle:** `UserSecretsClient` fails silently if Kaggle Settings → Internet is OFF. Documented prominently in notebook headers.
- GitHub contributor hygiene:** Early commits included Co-Authored-By: Claude trailers, causing @claude to appear in contributor graphs. **Fix:** amended commits with `git commit --amend` removing trailers; recreated repo when cached data persisted.

12. Repository Structure

├── README.md # Setup + all live links

```
|— .gitignore          # Excludes data/, results/, .env

└— docs/              # Internal planning docs (not graded)

    |— TEAM_DECISION.md # Model-selection voting doc

    |— PLAN_v2.md      # Execution plan with role
assignments

    └— REPORT.md       # This file
```